

Artificial Intelligence

AI2002

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GENETIC ALGORITHM - Introduction

- A biologically inspired model of intelligence
- The principles of biological evolution are applied to find solutions to difficult problems
- The problems are not solved by reasoning logically about them; rather populations of competing candidate solutions are spawned and then evolved to become better solutions through a process patterned after biological evolution

GENETIC ALGORITHM - Introduction

- Less worthy candidate solutions tend to die out, while those that show promise of solving a problem survive and reproduce by constructing new solutions out of their components
- GA begin with a **population of candidate problem solutions**
- Candidate solutions are evaluated according to their ability to solve problem instances: **only the fittest survive and combine with each other to produce the next generation of possible solutions**

GENETIC ALGORITHM - Introduction

- Thus increasingly powerful solutions emerge in a Darwinian universe. Learning is viewed as a competition among a population of evolving candidate problem solutions
- This method is heuristic in nature and it was introduced by John Holland in 1975

GENETIC ALGORITHM - Basic Algorithm

```
begin
  set time  $t = 0$ ;
  initialise population  $P(t) = \{x_1^t, x_2^t, \dots, x_n^t\}$  of solutions;
  while the termination condition is not met do
    begin
      evaluate fitness of each member of  $P(t)$ ;
      select some members of  $P(t)$  for creating offspring;
      produce offspring by genetic operators;
      replace some members with the new offspring;
      set time  $t = t + 1$ ;
    end
  end
```

GENETIC ALGORITHM - Basic Attributes of Genetic Algorithm

- Genetic representation of candidate solutions (Chromosome)
- Population size
- Fitness (Evaluation) function
- Genetic Operators
- Selection technique/algorithm
- Generation gap
- Amount of elitism used
- Number of duplicates allowed

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

- Gene: A basic unit, which represents one characteristic of the individual. The value of each gene is called an allele
- Chromosome: A string of genes; it represents an individual i.e. a possible solution of a problem. Each chromosome represents a point in the search space
- Population: A collection of chromosomes
- An appropriate chromosome representation is important for the efficiency and complexity of the GA

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

- The classical representation scheme for chromosomes is binary vectors of fixed length. However, real valued chromosomes are also possible
- In **binary** coded chromosomes, **every gene has two alleles**
- In **real** coded chromosomes, **a gene can be assigned any value from a domain of values**
- In the case of an **n-dimensional search space**, each **chromosome** consists of **n variables** with **each variable encoded as a bit string**

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

- Example: Cookies Problem
- Two parameters sugar and flour (in kgs). The range for both is 0 to 9 kgs. Therefore a chromosome will comprise of two genes called sugar and flour

5	1
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Chromosome # 01

2	4
---	---

Chromosome # 02

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

Example: Expression satisfaction Problem

$$\begin{aligned} F = & (\neg a \vee c) \wedge (\neg a \vee c \vee \neg e) \\ & \wedge (\neg b \vee c \vee d \vee \neg e) \wedge (a \vee \neg b \vee c) \\ & \wedge (\neg e \vee f) \end{aligned}$$

Chromosome: Six binary genes a b c d e f e.g. 100111

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

Example: Model Learning

Use GA to learn the concept Yes Reaction from the Food Allergy problem's data

Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
2	Lobdell	lunch	Friday	expensive	no
3	Sam's	lunch	Saturday	cheap	yes
4	Sarah's	breakfast	Sunday	cheap	no
5	Sam's	breakfast	Sunday	expensive	no

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

Example: Model Learning

A potential model of the data can be represented as a chromosome with the genetic representation:

Gene # 1 Restaurant	Gene # 2 Meal	Gene # 3 Day	Gene # 4 Cost
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The alleles of genes are:

Restaurant gene: Sam, Lobdell, Sarah

Meal gene: breakfast, lunch

Day gene: Friday, Saturday, Sunday

Cost gene: cheap, expensive

GENETIC ALGORITHM - Representation of Solutions: The Chromosome

Example

Find a number: 001010

You have guessed this binary number. If you write a program to find it, then there are $2^6 = 64$ possibilities

If you find it with the help of Genetic Algorithm, then the program gives a number and you tell its fitness

The fitness score is the number of correctly guessed bits

GENETIC ALGORITHM

Example

Find a number: 001010

Step 1. Chromosomes produced.

A) 010101	-	1
B) 111101	-	1
C) 011011	-	4 *
D) 101100	-	3 *

Best ones C & D

GENETIC ALGORITHM

Example

Find a number: 001010

Mating	New Variants	Evaluation
C) 01:1011	01:1100 (E)	3
D) 10:1100	10:1011 (F)	4 *
C) 0110:11	0110:00 (G)	4 *
D) 1011:00	1011:11 (H)	3

Selection of F & G

GENETIC ALGORITHM

Example

Mating	New Variants	Evaluation
F) 1:01011	1:11000 (H)	3
G) 0:11000	0:01011 (I)	5 *
F) 101:011	101:000 (J)	4 *
G) 011:000	011:011 (K)	4

Selection of I and J

GENETIC ALGORITHM

Example

	Mating	New Variants	Evaluation
I)	0010:11	0010:00 (L)	5
J)	1010:00	1010:11 (M)	4
I)	00101:1	00101:0 (N)	6 (success) *
J)	10100:0	10100:1 (O)	3

In this game success was achieved after 16 questions, which is 4 times faster then checking all possible $2^6 = 64$ combination